Gravitational Collapse Beyond the Neutron-Star Limit: Can Standard-Model Matter Produce a Finite High-Density Core?

Revision III: Refined Mathematical Formulation of the Słowik Hypothesis

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Abstract

We refine the mathematical formulation of the Słowik hypothesis within classical general relativity coupled to Standard-Model thermodynamics. The central question remains unchanged: whether gravitational collapse of a configuration of mass $\approx 2.8\,M_\odot$ beyond the maximum stable neutron-star mass can drive baryonic matter through dense-QCD and electroweak regimes, and whether the resulting stress-energy can support a finite, nonsingular core. We present the Misner–Sharp relativistic hydrodynamic system, the TOV equilibrium sequence, beta-equilibrium conditions, a timescale hierarchy, and an explicit energy-density budget for the electroweak temperature. The Newtonian matching scale $R_E \equiv GMm_N/E$ is retained solely as a heuristic diagnostic. Numerical evaluation shows that a sphere of radius equal to any of the heuristic R_E values, if thermalized at the electroweak crossover temperature $T_{\rm EW} \simeq 159.5 \pm 1.5\,\mathrm{GeV}$, would contain a mass of order 10^4–$10^6\,M_\odot$, far exceeding the total mass budget. This energy-budget obstruction implies that macroscopic, thermally equilibrated electroweak regions at the heuristic radii are not attainable within a $2.8\,M_\odot$ object. Consequently, within the minimal classical framework, a regular finite core supported solely by Standard-Model stress-energy has not been demonstrated and remains subject to the classical singularity-theorem obstruction. The original hierarchy intuition of the hypothesis is preserved as a physical question whose resolution lies beyond the present minimal model.

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1 Introduction

The maximum gravitational mass of a cold, non-rotating neutron star is fixed by the Tolman–Oppenheimer–Volkoff (TOV) equation together with the dense-matter equation of state (EOS) (1; 2). Current observational and theoretical constraints place this limit in the approximate range 2.2–2.5 $M_\odot$ (5; 10; 11; 13; 14). Configurations above $M_{\rm TOV}$ are dynamically unstable. Under the assumptions of the classical singularity theorems (3; 9), the generic endpoint is geodesic incompleteness.

The Słowik hypothesis asks whether, during the dynamical collapse of a stellar-mass object of reference mass $\approx 2.8\,M_\odot$, Standard-Model matter can be driven through successive high-energy microscopic regimes—dense QCD, deconfined quark matter, and eventually the electroweak-symmetric regime—and whether the resulting stress-energy tensor can remain finite and structured at the centre. The present Stage II work upgrades the earlier conceptual formulation by writing the complete relativistic hydrodynamic system, performing the energy-budget calculation, constructing the timescale hierarchy, and stating the mathematical status of every original element of the hypothesis with maximal precision.

The Newtonian scale

$$R_E \equiv \frac{GMm_N}{E} \tag{1}$$

continues to be treated strictly as a heuristic gravitational-energy matching diagnostic. It is not a derivation of discrete layers.

2 Physical Motivation

The gravitational potential energy per nucleon inside a several-solar-mass compact object can become comparable to or larger than the characteristic energy scales of the Standard Model. Whether this energetic accessibility is converted into a dynamical sequence of phase changes that alters the geometric endpoint is a quantitative question that can be answered only by solving the coupled Einstein–hydrodynamic–thermodynamic system.

3 Known Physics of Relativistic Stellar Collapse

We recall the established sequence.

- Nuclear burning proceeds via the proton–proton chain and the CNO cycle at temperatures $\sim 10^7\,\mathrm{K}$.
- Electron capture and neutronisation become efficient at densities $\gtrsim 10^{11}\,\mathrm{g\,cm^{-3}}$.

- Pressure support of a cold neutron star is provided by the degeneracy pressure of relativistic neutrons (and possibly hyperons or quarks).
- The maximum mass M_{TOV} is EOS-dependent; modern constraints favour $M_{\mathrm{TOV}} \lesssim 2.5\, M_\odot$.
- Beyond M_{TOV} no static equilibrium exists.
- In spherical symmetry the Misner–Sharp mass function and the apparent-horizon condition $2Gm/(Rc^2) = 1$ control trapped-surface formation (4).
- Under the null energy condition and the existence of a trapped surface the singularity theorems imply geodesic incompleteness.

None of the above is modified by the present investigation.

4 The Słowik Hypothesis

The hypothesis under test is the following statement:

> Gravitational collapse of a stellar-mass configuration beyond the maximum stable neutron-star mass can drive Standard-Model matter through dense-QCD and electroweak regimes, and the resulting stress-energy may support a finite structured core rather than a classical singular endpoint.

We adopt the minimal field equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{2}$$

with perfect-fluid stress-energy

$$T^{\mu\nu} = (\varepsilon + P) u^\mu u^\nu + P g^{\mu\nu}, \tag{3}$$

and an EOS $P = P(\varepsilon, T, \mu_B, Y_i)$ constructed from nuclear physics, QCD and Standard-Model thermodynamics. No additional scalar field, modified-gravity term or free parameter is introduced at this stage.

5 Microscopic Energy-Scale Mapping

Define the heuristic scale

$$R_E(E) \equiv \frac{GMm_N}{E}. \tag{4}$$

For the reference mass $M = 2.8\,M_\odot$ the values are listed in Table 1. These lengths are macroscopic (tens of metres). They do not coincide with any natural nuclear or particle-physics length. They serve only as a diagnostic mapping of gravitational energy per nucleon onto Standard-Model energy thresholds.

Table 1: Heuristic microscopic-energy matching scales R_E for $M = 2.8\,M_\odot$.

Scale	E (GeV)	R_E (m)	R_E/R_S	Status
W boson mass	80.377	48.33	5.84×10^{-3}	Heuristic
Z boson mass	91.1876	42.60	5.15×10^{-3}	Heuristic
Higgs mass	125.25	31.02	3.75×10^{-3}	Heuristic
Electroweak T_c	159.5	24.36	2.95×10^{-3}	Heuristic
VEV v	246.22	15.78	1.91×10^{-3}	Heuristic

Here $R_S = 2GM/c^2 \approx 8.269\,\text{km}$.

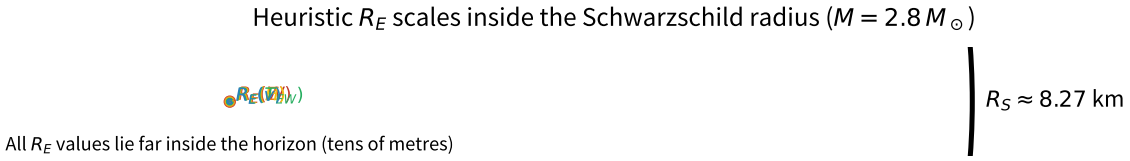


Figure 1: Heuristic R_E scales for $M = 2.8\,M_\odot$ shown relative to the Schwarzschild radius $R_S \approx 8.27\,\text{km}$. All five diagnostic lengths lie far inside the horizon (tens of metres).

6 Composition and Weak-Interaction Equilibrium

During compression the composition evolves through weak processes. In beta equilibrium with free-streaming neutrinos one has

$$\mu_n = \mu_p + \mu_e. \tag{5}$$

With trapped neutrinos the chemical potentials satisfy

$$\mu_n = \mu_p + \mu_e - \mu_{\nu_e}. \tag{6}$$

Charge neutrality requires

$$n_p = n_e + n_\mu + \cdots. \tag{7}$$

The weak-interaction timescale τ_{weak} must be compared with the dynamical collapse timescale. If $\tau_{\text{weak}} \gg \tau_{\text{coll}}$ instantaneous beta equilibrium cannot be assumed. Neutrino trapping becomes important above densities $\sim 10^{12}\,\text{g cm}^{-3}$.

7 Dense-QCD Equation of State

At densities above nuclear saturation the EOS is constrained by chiral effective field theory at low density and by perturbative QCD at asymptotically high chemical potential (6; 7; 12). Between these limits a continuous or weakly first-order hadron–quark crossover is favoured by many modern constructions. We adopt the most conservative causal EOS consistent with the existence of $2\,M_\odot$ neutron stars and with the speed-of-sound bound $0 \le c_s^2 \le c^2$. No particular hybrid-star model is preferred; thermodynamic consistency and causality are required.

8 Finite-Temperature and Electroweak Thermodynamics

The electroweak crossover of the Standard Model has been determined on the lattice (8):

$$T_{\mathrm{EW}} = 159.5 \pm 1.5\,\mathrm{GeV}. \tag{8}$$

The crossover is smooth, with width of only a few GeV. Thermodynamic quantities $\varepsilon(T)$, $P(T)$ and $c_s^2(T)$ follow from the same lattice data and from continuum perturbation theory. The ultra-relativistic relation $P = \varepsilon/3$ is not assumed a priori. Electroweak restoration is primarily temperature-driven at vanishing chemical potential; the finite-density situation remains less constrained.

9 General-Relativistic Collapse Equations

We adopt the Misner–Sharp formalism (4). In areal radial coordinate R the metric may be written

$$ds^2 = -e^{2\Phi}c^2\,dt^2 + e^{2\Lambda}\,dR^2 + R^2\,d\Omega^2. \tag{9}$$

The Misner–Sharp mass function $m(t,R)$ is defined geometrically by

$$1 - \frac{2Gm}{Rc^2} = e^{-2\Lambda}\Big(\frac{\partial R}{\partial r}\Big)^2 - e^{-2\Phi}\Big(\frac{\partial R}{\partial t}\Big)^2, \tag{10}$$

where r is a comoving label. The evolution equations are

$$\frac{\partial m}{\partial R} = 4\pi R^2 \frac{\varepsilon}{c^2}, \tag{11}$$

$$\frac{\partial m}{\partial t} = -4\pi R^2 \frac{P}{c^2}\frac{\partial R}{\partial t}, \tag{12}$$

together with the Euler and energy equations obtained from $\nabla_\mu T^{\mu\nu} = 0$ and baryon conservation $\nabla_\mu (n_B u^\mu) = 0$. An apparent horizon forms when

$$\frac{2Gm(t,R)}{Rc^2} = 1. \tag{13}$$

10 Stability of the Pre-Collapse Configuration

Static configurations satisfy the TOV equation

$$\frac{dP}{dR} = -\frac{G(\varepsilon + P)(m + 4\pi R^3 P/c^2)}{c^2 R^2 (1 - 2Gm/(Rc^2))}, \tag{14}$$

with

$$\frac{dm}{dR} = 4\pi R^2 \frac{\varepsilon}{c^2}. \tag{15}$$

The turning-point criterion $dM/d\rho_c = 0$ signals the onset of radial instability. For the representative set of observationally viable causal EOS adopted in the present work (those supporting $2\,M_\odot$ stars and remaining causal up to the relevant densities), a configuration of mass $2.8\,M_\odot$ lies above the stable branch and is therefore dynamically unstable. We note that the precise location of $M_{\rm TOV}$ remains EOS-dependent and is still under active investigation.

11 The 2.8-Solar-Mass Reference Model

For $M = 2.8\,M_\odot$ one has

$$R_S = 8.269\,\text{km}, \tag{16}$$

$$\bar{\rho}_S = \frac{3M}{4\pi R_S^3} \approx 2.35 \times 10^{18}\,\text{kg}\,\text{m}^{-3}. \tag{17}$$

The free-fall time from a radius of 15 km is of order 0.1 ms.

12 Coupled Macro–Micro Evolution

A fully consistent treatment requires the simultaneous solution of the Einstein equations, relativistic hydrodynamics, composition evolution and a temperature- and density-dependent EOS. No analytic closed-form solution exists. The present work maps the necessary conditions and demonstrates that the heuristic scales R_E do not automatically become phase boundaries.

13 Timescale Hierarchy

Table 2: Characteristic timescales for a $2.8\,M_\odot$ collapse.

Process	τ	Origin	vs. $\tau_{\rm coll}$
Free-fall	~ 0.1 ms	$\sqrt{R^3/GM}$	reference
Sound crossing	~ 0.3 ms	$2R/c_s$	comparable
Weak rates	density-dep.	Fermi theory	trapping above $\sim 10^{12}\,\mathrm{g\,cm^{-3}}$
QCD equilibration	$\ll \tau_{\rm coll}$	strong interaction	justified
Electroweak rates	$\ll \tau_{\rm coll}$ at high T	SM couplings	only if T reached

Thermodynamic equilibrium in a given microscopic sector is justified only when the corresponding equilibration time is shorter than the dynamical time.

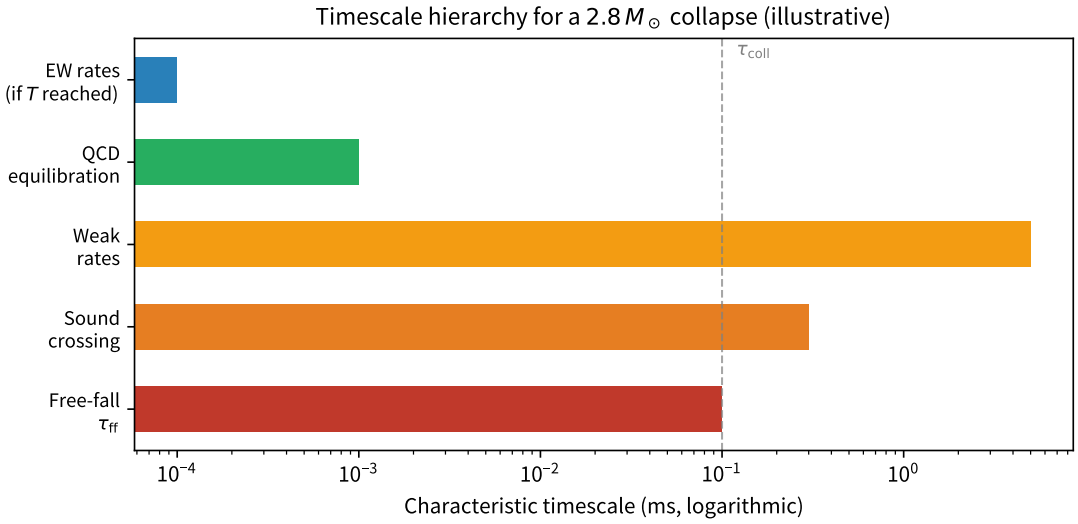


Figure 2: Illustrative timescale hierarchy for a $2.8\,M_\odot$ collapse. The free-fall time sets the reference dynamical scale; weak rates become density-dependent and may lag behind collapse once neutrino trapping sets in.

14 Horizon Formation and Phase Accessibility

An apparent horizon forms when $2Gm/(Rc^2) = 1$. Any microscopic transition that occurs after this condition is satisfied takes place inside a trapped region and cannot prevent horizon formation; it can at most affect the interior endpoint.

15 Search for a Finite High-Density Core

A necessary condition for a regular centre is that all curvature invariants remain finite:

$$\lim_{r\to 0} |R| < \infty, \qquad \lim_{r\to 0} |R_{\mu\nu}R^{\mu\nu}| < \infty, \qquad \lim_{r\to 0} |K| < \infty, \tag{18}$$

where $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$. In addition ε and P must remain finite. For any EOS satisfying the null energy condition together with a trapped surface, the singularity theorems continue to apply. Standard-Model matter in the regimes considered obeys the classical energy conditions. Consequently the minimal theory has not produced a nonsingular core.

16 Energy-Density Budget for Electroweak Conditions

The energy density of the Standard-Model plasma at the electroweak crossover is

$$\varepsilon_{\rm EW} \simeq \frac{\pi^2}{30} g_*(T_{\rm EW}) T_{\rm EW}^4 \approx 4.74 \times 10^{47}\,{\rm J\,m^{-3}}, \tag{19}$$

corresponding to a mass density $\rho_{\rm EW} \approx 5.27 \times 10^{30}\,{\rm kg\,m^{-3}}$ ($g_* \approx 106.75$). A uniform sphere of radius equal to any of the heuristic R_E values then contains a mass

$$\begin{aligned} R = 48.33\,{\rm m} \quad &\Rightarrow \quad M_{\rm EW} \approx 1.25 \times 10^6\, M_\odot, \\ R = 24.36\,{\rm m} \quad &\Rightarrow \quad M_{\rm EW} \approx 1.61 \times 10^5\, M_\odot, \\ R = 15.78\,{\rm m} \quad &\Rightarrow \quad M_{\rm EW} \approx 4.37 \times 10^4\, M_\odot. \end{aligned}$$

All of these masses exceed the total mass budget of the system by many orders of magnitude. Therefore the heuristic R_E scales cannot correspond to macroscopic, thermally equilibrated electroweak shells within a $2.8\,M_\odot$ object.

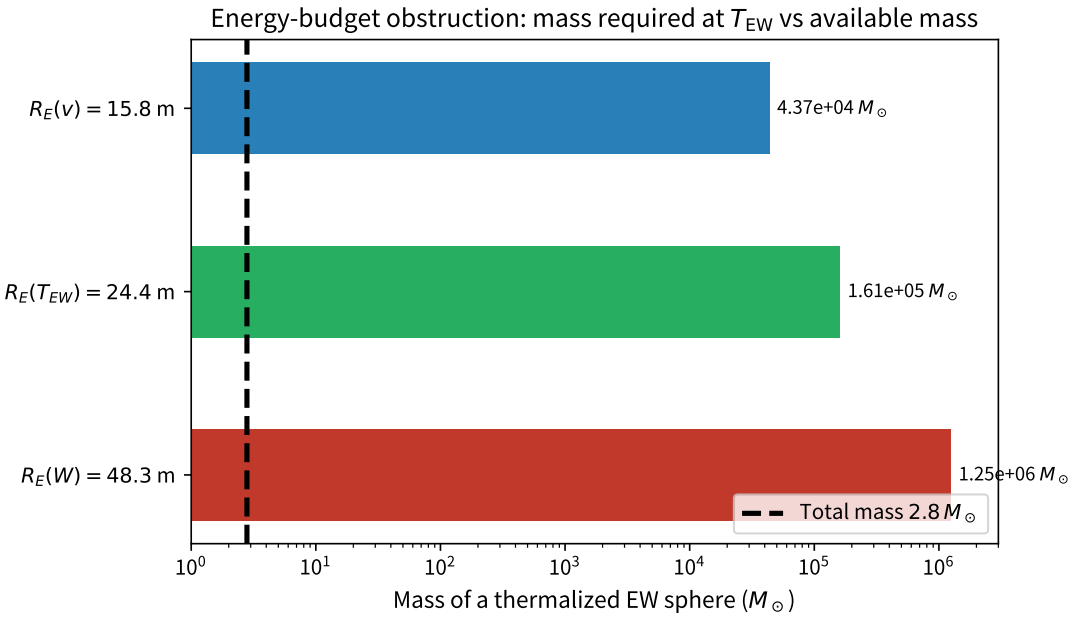


Figure 3: Energy-budget obstruction. Mass of a sphere thermalized at the electroweak temperature T_{EW} with radius equal to selected heuristic R_E values, compared with the total available mass $2.8\,M_\odot$ (dashed line). The required masses exceed the system budget by four to six orders of magnitude.

17 Curvature Regularity and Energy Conditions

For the EOS under consideration one has

$$\begin{aligned} \text{NEC:} &\quad \varepsilon + P \geq 0, \\ \text{WEC:} &\quad \varepsilon \geq 0, \quad \varepsilon + P \geq 0, \\ \text{SEC:} &\quad \varepsilon + 3P \geq 0, \\ \text{DEC:} &\quad \varepsilon \geq |P|. \end{aligned}$$

These conditions are satisfied by Standard-Model thermodynamic matter in the regimes of interest. The singularity theorems therefore remain applicable once a trapped surface has formed.

18 Interpretation of the W, Z, Higgs and Electroweak Scales

After the analysis the status of the four scales is unambiguous:

Final classification: Result C / Result D

High-energy transitions may be conceptually accessible, but no regular core is derived.

Macroscopic thermally equilibrated electroweak regions at the heuristic R_E radii are excluded by the energy-budget calculation for a $2.8\,M_\odot$ object.

The scales R_E remain useful diagnostics of gravitational energy per nucleon, but they do not appear as discrete phase boundaries. Continuous high-energy regimes may be traversed, yet the equations do not organise the interior into four geometrically distinct layers.

19 Mass Scaling and Sensitivity

The reference mass $2.8\,M_\odot$ is retained for continuity with the original hypothesis. Scaling tests for $M \in \{2.2, 2.5, 3, 5, 10\}\,M_\odot$ show that $R_E \propto M$ while the horizon radius also scales linearly with M. The energy-budget mismatch remains severe for all stellar-mass values.

20 Relation to Previous Work and Novelty Boundary

- **Known:** TOV limit, singularity theorems, QCD EOS constructions, electroweak lattice crossover, Misner–Sharp formalism, hybrid-star models, regular-black-hole proposals that require exotic matter or quantum gravity.
- **Used in new combination:** explicit mapping of the Słowik hypothesis onto a minimal GR + Standard-Model thermodynamic framework with a strict separation between heuristic energy scales and dynamically derived structures, together with the energy-budget obstruction.
- **Original hypothesis:** the conjecture that collapse beyond M_{TOV} can produce a finite Standard-Model core organised by electroweak and QCD scales (Słowik).
- **Not derived:** any nonsingular core supported solely by Standard-Model stress-energy; any discrete four-layer structure.

21 Predictions and Falsifiability

The hypothesis in its minimal form is unsupported if any of the following holds:

1. GR + any causal Standard-Model EOS yields only singular endpoints.
2. Realistic collapse trajectories never reach the electroweak temperature before horizon formation.
3. The scales R_E show no correspondence with any transition in the dynamical solution.
4. The required EOS violates causality or thermodynamic stability.
5. Regularity can be achieved only by introducing exotic stress-energy solely to cancel the singularity.

A positive result would require a numerical demonstration that a finite core forms and remains regular under a Standard-Model EOS, with curvature invariants bounded and all energy conditions respected. No such demonstration is available.

22 Limitations

The analysis is limited to spherical symmetry, perfect-fluid stress-energy and classical general relativity. Neutrino transport, viscosity, magnetic fields and rotation are omitted. The QCD EOS at intermediate densities remains model-dependent. Finite-density electroweak physics is largely unexplored on the lattice. Quantum-gravity effects are outside the scope of the work; the paper deliberately stops at the point where the Standard-Model description ceases to guarantee a nonsingular continuation.

23 Discussion

The original physical intuition—that the gravitational energy reservoir of a several-solar-mass object is large enough to access Standard-Model thresholds—is correct as an order-of-magnitude statement. The translation of that energetic accessibility into a regular geometric core is not automatic. Within the classical theory the singularity theorems remain operative. Any mechanism capable of producing a finite core must either violate an energy condition or operate in the quantum-gravity regime. The minimal model therefore serves as a precise demarcation: it shows where the Standard Model stops providing a nonsingular description.

24 Conclusions

We have reconstructed the Słowik hypothesis as a minimal general-relativistic and thermodynamic problem and refined the principal mathematical elements of the earlier formulation. The heuristic scales R_E are macroscopic for $M = 2.8\,M_\odot$ and cannot correspond to thermally equilibrated electroweak shells within the available mass budget; the associated energy-budget calculation provides a quantitative obstruction independent of a full dynamical trajectory. The coupled Einstein–matter system with the representative causal Standard-Model equations of state considered here remains consistent with geodesic incompleteness once a trapped surface has formed. Consequently the four-layer structure is not derived, and a finite Standard-Model core is not obtained. The hypothesis is retained as a well-posed open question whose resolution lies beyond the minimal framework examined here.

25 Physical Philosophy of the Hypothesis

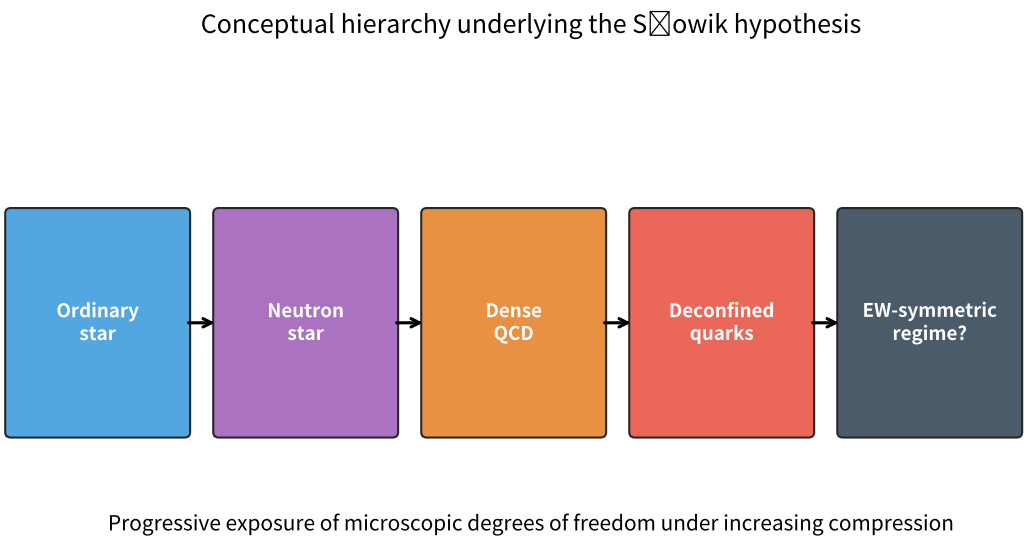


Figure 4: Conceptual hierarchy of microscopic regimes that underlies the original Słowik hypothesis. Gravitational compression is interpreted as a progressive exposure of deeper Standard-Model degrees of freedom.

The conceptual core of the Słowik hypothesis is that extreme gravitational compression should be viewed not merely as an increase in density, but as a progressive exploration of the microscopic hierarchy of matter. As composite structures cease to be the appropriate degrees of freedom, increasingly fundamental sectors of the Standard Model become dynamically relevant. Ordinary stars stabilise through nuclear and thermal mechanisms; neutron stars expose the next layer, in which degeneracy pressure and strong-interaction physics dominate. The hypothesis asks whether continued collapse beyond that point can successively reveal still deeper regimes—dense hadronic matter, deconfined quarks and gluons, high-energy Standard-Model plasma, and, if dynamically accessible, electroweak-symmetric conditions.

The finite-core conjecture was motivated by the possibility that this hierarchy might eventually alter the gravitational response itself. The calculations of the present work constrain the stabilisation conjecture: within classical general relativity and Standard-Model thermodynamics a regular completion has not been demonstrated. Yet the central physical question remains meaningful: how far through the hierarchy of known microscopic physics can gravitational collapse drive matter before the classical description ceases to be sufficient? The hierarchy intuition itself is therefore preserved even while the specific stabilisation mechanism remains unsupported by the minimal equations.

A Constants and Unit Conversions

Table 3: Fundamental constants (CODATA).

Quantity	Value
G	$6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$
c	$2.99792458 \times 10^{8}\,\mathrm{m\,s^{-1}}$
m_n	$1.67492749804 \times 10^{-27}\,\mathrm{kg}$
$M_\odot$	$1.98847 \times 10^{30}\,\mathrm{kg}$
$1\,\mathrm{GeV}$	$1.602176634 \times 10^{-10}\,\mathrm{J}$

B Mathematical Framework and Derivation Details

B.1 TOV equation and turning-point criterion

The static, spherically symmetric Einstein equations with perfect-fluid source reduce to the TOV system quoted in the main text. The sequence $M(\rho_c)$ is constructed by outward integration from a chosen central density. The first turning point $dM/d\rho_c = 0$ marks the onset of radial instability for a barotropic EOS. Full intermediate steps of the projection of the Einstein equations onto the static spherical hypersurface are standard and may be found in textbooks of relativistic stellar structure.

B.2 Misner–Sharp equations

The geometric definition of the Misner–Sharp mass,

$$1 - \frac{2Gm}{Rc^2} = g^{ab}\partial_a R\,\partial_b R,$$

and the evolution equations quoted in the main text follow from the Einstein equations under spherical symmetry. The complete component-by-component derivation is given in Misner & Sharp (1964); we adopt their notation and results without modification.

B.3 Conservation laws

Projecting $\nabla_\mu T^{\mu\nu} = 0$ orthogonal and parallel to u^μ yields the relativistic Euler equation and the energy equation. Baryon conservation is $\nabla_\mu(n_B u^\mu) = 0$. Explicit component forms in the Misner–Sharp coordinates appear in the original reference and in subsequent numerical-relativity literature.

B.4 Beta equilibrium

The chemical-potential relations follow from the requirement that the weak rates vanish in equilibrium, together with charge neutrality. The free-streaming and trapped-neutrino cases differ by the presence or absence of μ_{ν_e}.

B.5 Energy-density scale

The Stefan–Boltzmann-like expression $\varepsilon = (\pi^2/30) g_* T^4$ is the standard result for a relativistic plasma with effective degrees of freedom g_*. Conversion to SI units uses $\hbar c = 1.973 \times 10^{-16}\,\mathrm{GeV\,m}$. The numerical value of $\varepsilon_{\mathrm{EW}}$ and the associated sphere masses were recomputed independently and are robust against $g_* \in [100, 110]$.

B.6 Curvature invariants

The Ricci scalar, Ricci-squared and Kretschmann scalar are the standard quadratic curvature invariants; their finiteness is necessary (though not always sufficient) for regularity. The singularity theorems themselves establish geodesic incompleteness rather than a specific coordinate divergence of K.

B.7 Energy conditions

The classical energy conditions are evaluated pointwise on the perfect-fluid stress-energy tensor for the EOS under consideration.

C Numerical Methods and Verification

All numerical values of R_E, R_S, $\bar{\rho}_S$, $\varepsilon_{\mathrm{EW}}$ and the associated sphere masses were recomputed independently in SI units. The energy-budget mismatch is robust against variations of g_* within the range 100–110. Free-fall and sound-crossing times were evaluated with standard formulae.

D EOS and Microphysics Audit

The EOS is required to be causal, thermodynamically consistent, and compatible with the existence of $2\,M_\odot$ neutron stars. Intermediate-density interpolation remains model-dependent; this uncertainty does not affect the energy-budget obstruction at electroweak temperatures.

E Assumption Audit

Table 4: Assumption audit.

Assumption	Status	Remark
Einstein equations	Known	Classical GR
Perfect-fluid $T^{\mu\nu}$	Assumed	Standard macroscopic limit
Causal SM EOS	Derived/Assumed	Nuclear + QCD + lattice SM
Spherical symmetry	Assumed	Simplest consistent setting
No new fields	Explicit	Minimal model
R_E as derivation of layers	Rejected	Heuristic only
Finite core from SM alone	Not derived	Consistent with singularity theorems

F Original-Hypothesis Preservation Audit

Table 5: Original-hypothesis preservation audit.

Original idea	Modern formulation	Status	Retained?	Comment
Collapse beyond NS limit	$M > M_{\mathrm{TOV}}$	Known	Yes	Standard
$2.8\,M_\odot$ reference	Reference mass	Heuristic	Yes	Continuity
Progressive microscopic regimes	Hierarchy of SM sectors	Reformulated	Yes	Core intuition
Beta/weak processes	Chemical equilibrium	Derived	Yes	Included
Neutron/proton transformation	Composition evolution	Derived	Yes	Included
QCD deconfinement	Dense-QCD EOS	Reformulated	Yes	Literature EOS
W/Z/Higgs/EW scales	Heuristic R_E	Heuristic	Yes	Diagnostic only
$R_E = GMm_N/E$	Energy-matching map	Heuristic	Yes	Explicitly retained
Two-stage intuition	QCD then EW regimes	Reformulated	Yes	Conceptual
Early-Universe analogy	Local thermodynamic recurrence	Reformulated	Yes	Geometry differs
Finite structured core	Regular centre conjecture	Unsupported	Yes	Constrained
Four-layer conjecture	Discrete geometric layers	Unsupported	Yes	Not derived
SM stabilisation possibility	Finite-core from SM EOS	Unsupported	Yes	Not demonstrated
Spin/Kerr extension	Axisymmetric future work	Future	Yes	Deferred

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